TECHNICAL DATABASE SCHEMA MIGRATION

Schema Migration Plan: User Profiles v2

DATABASE SCHEMA MIGRATION PLAN

Target Database: `meridian_auth_prod` Schema Version: Upgrading from v14.2 to v15.0 Execution Window: August 22, 2026 (02:00 - 04:00 UTC) DBA Lead: Sarah Jenkins

Objective: The upcoming release requires flattening the `user_preferences` JSONB blob into distinct, indexed relational columns to support the new high-speed matchmaking algorithm. This migration involves creating three new tables, copying 45 million rows, and dropping legacy constraints without causing a write-lock on the active user table.

Migration Strategy (Zero-Downtime Approach): We will utilize the Expand-and-Contract migration pattern to ensure backward compatibility for API clients during the rollout.

Phase 1: Expand (Pre-Deployment) 1. Create the new `user_settings`, `privacy_configs`, and `notification_prefs` tables. 2. Add dual-write triggers to the existing `users` table. Any new inserts or updates to the legacy JSONB blob will automatically be parsed and written to the new relational tables. 3. Execute a background backfill script to migrate the historical 45 million records into the new tables. This script will be throttled to 1,000 rows/second to prevent IOPS spiking.

Phase 2: Migrate (Application Deployment) Deploy the updated application code (v4.0). The new code will read from and write exclusively to the new relational tables, ignoring the legacy JSONB blob.

Phase 3: Contract (Post-Deployment Cleanup) Following a 48-hour observation period to ensure stability and zero data mismatch errors, we will execute the final schema contraction: 1. Drop the dual-write triggers. 2. Execute `ALTER TABLE users DROP COLUMN user_preferences;` to reclaim approximately 120GB of disk space.

Rollback Plan: If the backfill script causes query latency to exceed 500ms, the process will automatically abort. Application code can be quickly rolled back to v3.9, which will continue relying on the JSONB blob.